18 T is inversely proportional to the cube of u.

When $u = 5$, $T = 0.0096$

Find the value of u when $T = 0.15$

$T = k/u^3$; $k = Tu^3 = 0.0096 \times 5^3 = 1.2$.

1 mark
Find constant1.2

$0.15 = 1.2/u^3$, so $u^3 = 1.2 \div 0.15 = 8$.

1 mark
Rearrange for u^3

u = cube root of 8 = 2.

1 mark
u=2

(Total for Question 18 is 3 marks)